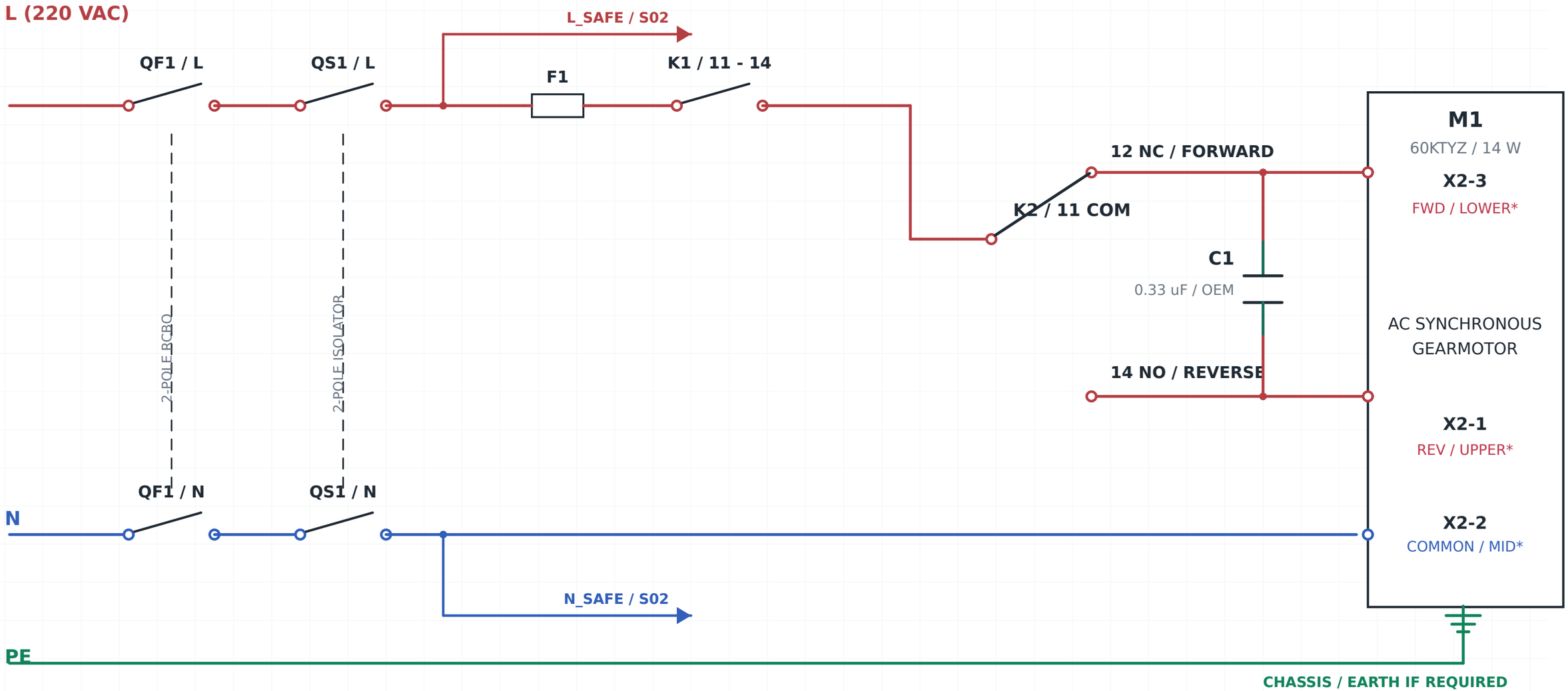
ESP32 SMART MIXER / ELECTRICAL SCHEMATIC

AC motor supply, winding selection and capacitor

IEC-style functional schematic | 220-230 VAC 50 Hz single-phase | NC = Forward / NO = Reverse



* Terminal layout inferred from YOUR photos; verify actual motor datasheet before energizing.

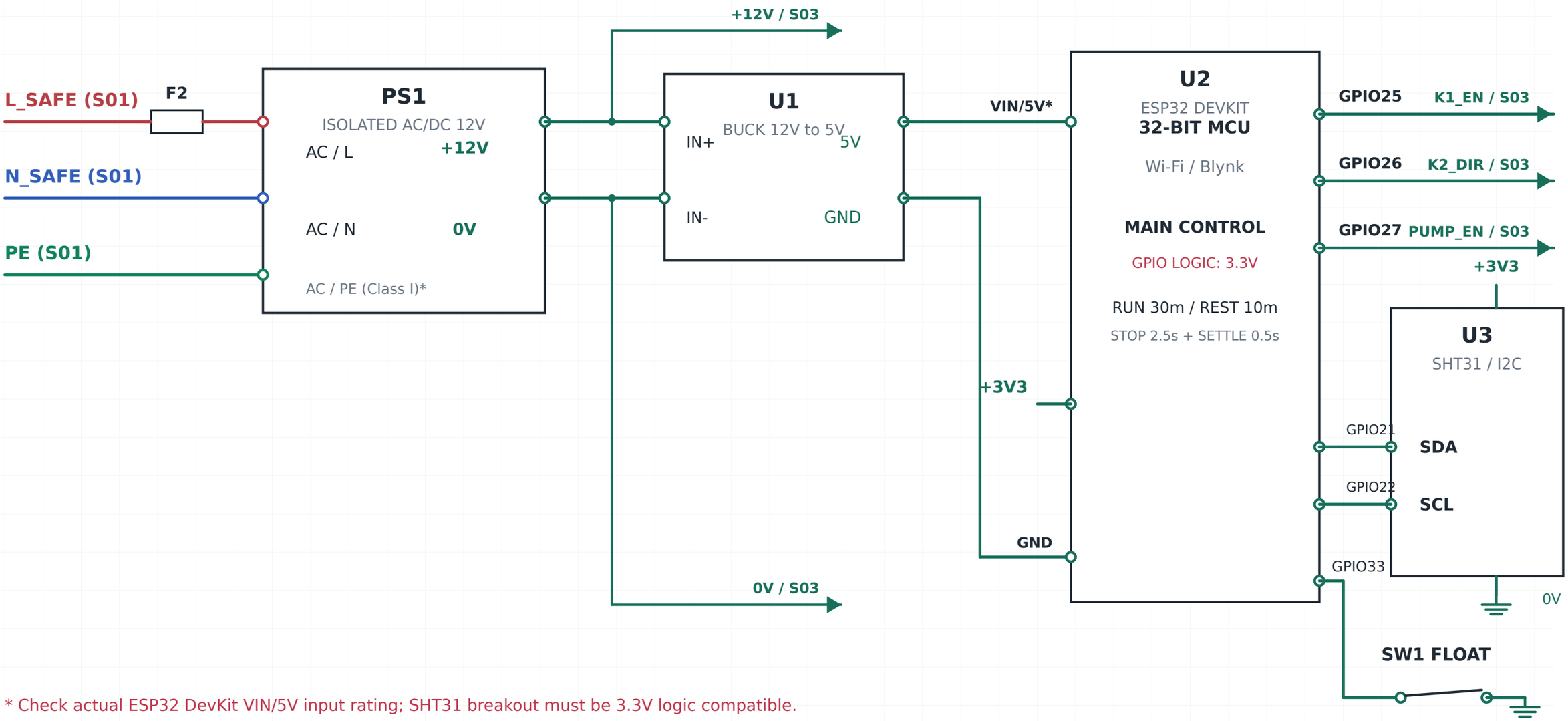
Keep original AC-rated motor capacitor. Check S1/S2 duty rating and motor-load torque.

DWG: ESP32-MIX-2026 STATUS: DESIGN REFERENCE / NOT SAFETY CERTIFIED	REV 3.0 | 11 NOV 2026 A3 LANDSCAPE | NOT TO SCALE	**SHEET 01 / 3** CHECK ALL PART RATINGS

ESP32 SMART MIXER / ELECTRICAL SCHEMATIC

Isolated DC power, ESP32 and SHT31 sensor

Net labels match sheets 01 and 03 | ESP32 3.3V I/O | No direct connection between AC N and DC 0V



* Check actual ESP32 DevKit VIN/5V input rating; SHT31 breakout must be 3.3V logic compatible.

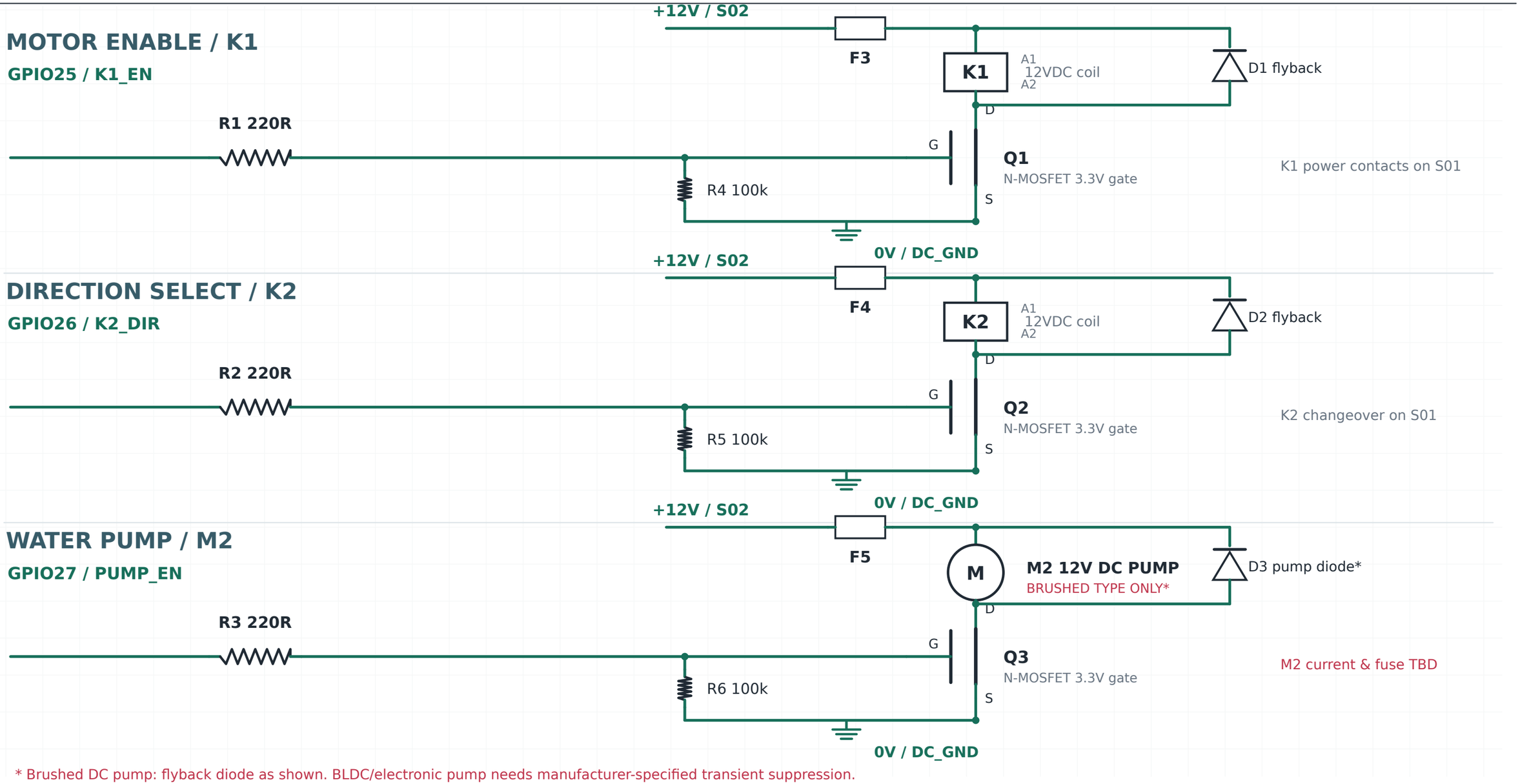
SHT31 measures air RH, not moisture percentage of the mixed material.

DWG: ESP32-MIX-2026 STATUS: DESIGN REFERENCE / NOT SAFETY CERTIFIED	REV 3.0 | 11 NOV 2026 A3 LANDSCAPE | NOT TO SCALE	**SHEET 02 / 3** CHECK ALL PART RATINGS

ESP32 SMART MIXER / ELECTRICAL SCHEMATIC

Discrete MOSFET outputs, relay coils and 12V pump

All GPIO outputs ACTIVE HIGH | K1 de-energized = motor OFF | K2 de-energized = FORWARD (NC)



* Brushed DC pump: flyback diode as shown. BLDC/electronic pump needs manufacturer-specified transient suppression.